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DEPARTMENT OF CSE

DATA SCIENCE

COURSE: 24AD3107

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TITLE OF THE PROJECT	COVID-19 / Public Health Data Visualization Dashboard	
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Abstract

The COVID-19 / Public Health Data Visualization Dashboard is a Data Science project developed to analyze COVID-19-related public health data and present the results in a simple and interactive way. The main aim of the project is to convert raw public health data into useful visual insights so that users can easily understand important trends and differences across time periods and regions.

First, we will collect a suitable COVID-19 or public health dataset from an open-data source. We will study the structure of the dataset and select the relevant attributes required for analysis. The data will then be cleaned and preprocessed by handling missing values, duplicate records, inconsistent values, and incorrect data formats wherever required. This step will make the dataset suitable for further analysis.

Next, we will perform Exploratory Data Analysis (EDA) to understand the data and identify important patterns and trends. We will use statistical summaries and comparisons to study how the available public health indicators change over different dates, regions, or categories. The purpose of EDA is to understand what is happening in the dataset before presenting the results visually.

After the analysis, we will create visualizations using Python libraries such as Matplotlib and Seaborn. Depending on the dataset, the dashboard will include line charts for trends over time, bar charts for comparisons, and distribution or other suitable plots for understanding the data. These visualizations will make the information easier to read and help users identify increases, decreases, and major differences without examining the raw dataset manually.

Finally, we will combine the processed data and visualizations into an interactive dashboard. The dashboard will provide filters such as region, category, or time period where applicable, allowing users to select the information they want to explore. The final output of the project will therefore be a user-friendly public health dashboard that follows the complete Data Science workflow: data collection, preprocessing, Exploratory Data Analysis, visualization, and interactive presentation.

Overall, this project demonstrates how Data Science and data visualization techniques can be used to understand a real-world public health dataset. Instead of only displaying raw numbers, the system presents the data as clear visual insights that can support better understanding and interpretation of COVID-19-related public health trends.

Signature of the Guide

HOD